PROJECT REFLOW: FINAL CASE COMPETITION SOLUTION

100-Point Scoring Rubric Defense & Executive Deliverable

DECISION: NorthStar Retail Group will execute Portfolio A+B, investing CAD $13.00M while preserving CAD $7.00M in capital reserve. Over the 3-year evaluation horizon, this initiative generates CAD $8.08M in annual net savings, produces +CAD $7.094M in discounted economic value (NPV), achieves simple payback in 19.3 months, and fully satisfies both non-financial mandates by dropping late deliveries to 3.5% and cutting emissions by 25.0%.

EXECUTIVE JUDGE STRESS-TEST DEFENSE SUMMARY

1. Why not Portfolio B+C (highest gross savings)?
   -> B+C carries $2.2M OpEx on C, making net savings ($8.22M) equal to A+B ($8.08M). However, B+C requires $17.0M Capex, crushing its 3-year NPV to $3.44M vs. $7.09M for A+B. A+B creates double the net shareholder value at lower risk.

2. What if implementation costs surge by 25%?
   -> Upfront investment increases from $13.0M to $16.25M, remaining safely below the $20.0M ceiling. Payback extends from 1.61 to 2.01 years (24.1 months), and NPV remains strongly accretive at +$3.84 Million.

3. What if projected annual savings fall 30% below plan?
   -> Net annual savings decline to $5.66M/yr. Payback increases to 2.30 years (27.6 months), still paying back within the 3-year window. Both service level (3.5%) and decarbonization (-25%) targets remain 100% intact.

4. How does strategy change if order volume surges 40%?
   -> Operating leverage expands dramatically. Robotics and algorithmic routing absorb volume with zero linear labor costs, pushing logistics cost per order below $5.90 and expanding annual net savings to $11.3M/yr (NPV > +$15.0M).

5. Why trust AI with mission-critical supply chain operations?
   -> Deterministic Human-in-the-Loop governance. Automated carrier routing includes fail-safe static dispatch fallbacks, and warehouse robotics operate as collaborative cobots assisting certified operators.